

Course Levels — From Clinical Question to Independent Research

Organisation: Neudata — #ClearDataClearImpact

Neudata's clinical-research training runs in three levels. Each level builds on the one before, and each department can stop at the level that fits its needs.

	Level 1 — Foundations	Level 2 — Doing the analysis	Level 3 — Advanced topics
In one line	Understand research from question to publication	Do the analysis yourself, properly, in R	Tackle complex designs and data with Neudata
Who	A whole department (~30 doctors, nurses, pharmacists)	An interested subgroup (8–12) who want to run their own analyses	Researchers with a specific complex project or dataset
Prerequisites	None	Level 1 + the free Neudata course <i>Clinical Data Analysis in R</i> (or equivalent basics)	Level 2 (or equivalent experience), plus a real question or dataset
Format	8 × 3 h over 4 weekends	8 × 3 h over about 4–6 weeks, with homework	Workshops of 1–2 sessions, or consultancy, on request
Software	None for participants (instructor demos)	R and RStudio on own laptops	R (and specialised packages)
Hands-on?	Concepts, reading and interpreting; analyses demonstrated live on FLUID-ICU	Every session is hands-on coding	Hands-on on the participant's own data
Outcome	A group research proposal and a group draft manuscript	A reproducible analysis and a draft Results section, on own or course data	A completed analysis plan or analysis for a real project, with Neudata support
Documents	Course/ (the full Level 1 course)	level2_hands_on.md	level3_advanced.md

How the levels connect

- **Level 1 → Level 2.** In Level 1, participants *read* the FLUID-ICU analyses (Table 1, crude vs adjusted ORs, Kaplan–Meier, multiple imputation) that the instructor runs. In Level 2 they **run, check and report those same analyses themselves**, then apply them to their own data. The FLUID-ICU dataset and R scripts in `Course/Data/` are shared by both levels.
- **Level 2 → Level 3.** Level 2 makes participants confident with descriptive statistics, linear and logistic regression, missing data and reproducible reporting. Level 3 takes the methods that real clinical data often demand — meta-analysis, prediction models, diagnostic accuracy, advanced survival analysis, longitudinal and multilevel models, large or clustered datasets — and works on them with the participant's own project.

Progression at a glance

Skill	Level 1	Level 2	Level 3
Research question, design, ethics	Taught in depth	Assumed	Assumed
Data management and cleaning	Concepts; live demo	Hands-on: import, clean, validate, cleaning log, data dictionary, version control	Large, linked or clustered data
Describing data, Table 1, figures	Read and judge	Build publication-ready tables and figures	Specialised plots (forest, funnel, calibration, cumulative incidence)
Regression	Read and interpret outputs	Fit, check, interpret and report linear and logistic models	Mixed models, competing risks, time-varying covariates, prediction models
Missing data	Mechanisms and ideas	Complete-case vs multiple imputation (mice) in practice	Complex imputation (multilevel, longitudinal)
Survival analysis	Read Kaplan–Meier and HRs	Intro only (via FLUID-ICU scripts)	In depth
Writing and publishing	Full manuscript workflow (group draft)	Reproducible report → Results section	Methods and results for complex analyses

Status: Level 1 materials are in [Course/](#). Level 2 and Level 3 are outlines — materials to be developed after Level 1 has run.